NLP-Enhanced Compatibility Modeling for Social Matching: A Text-First Hybrid Architecture Using the OKCupid Dataset

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Abstract

This paper presents a complete NLP-enhanced compatibility modeling system for social matching platforms, evaluated on self-description essay text and structured profile attributes from the OKCupid dataset ($N = 59{,}946$ profiles) (10). We evaluate three text representation methods—TF-IDF, Word2Vec average pooling, and SBERT sentence embeddings (6)— finding that SBERT is the only method producing rankable pairwise similarity distributions. A RoBERTa-based affect classifier (9) reveals a strong self-presentation bias in dating profiles, with a median positive sentiment probability of 0.855. A cross-signal correlation study ($n = 5{,}000$ pairs) demonstrates that text similarity and attribute compatibility are statistically independent ($r \approx 0.02$, $p > 0.1$), providing empirical justification for a hybrid fusion architecture. UMAP visualization of SBERT embeddings confirms that neither personality-compatible nor orientation-grouped profiles form detectable clusters in semantic space, corroborating the independence finding. A three-signal XGBoost re-ranker combining SBERT similarity, weighted attribute compatibility, and sentiment similarity is evaluated against a psychology-motivated compatibility label grounded in assortative personality matching (12), achieving AUC = 0.795 versus AUC = 0.793 for text similarity alone and AUC = 0.497 for attribute compatibility alone—demonstrating that text-derived signals dominate personality-based compatibility prediction while structured attributes provide orthogonal signal.

1 Introduction

Compatibility prediction in social matching platforms is conventionally driven by engagement signals—swipe rates, recency, and popularity—rather than measures of psychological or semantic alignment between users (1). Platforms such as Tinder and Hinge optimize for engagement metrics that favor popular profiles, creating a feedback loop that systematically disadvantages users whose compatibility is better predicted by personality or shared values than by profile popularity. This project investigates whether richer text-derived representations of user intent and personality can improve compatibility modeling, with the long-term goal of building a matching algorithm for a social app MVP that prioritizes genuine compatibility over surface-level engagement.

The core hypothesis is that user-generated free-text profile descriptions encode latent compatibility signals that behavioral signals alone cannot capture. Psychological research supports this hypothesis: De La Mare and Lee (12) demonstrate that online daters do take note of profile descriptions even when presented alongside photographs, and that people show assortative preferences for partners similar to themselves in openness and agreeableness—personality traits naturally expressed through

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self-descriptive writing. This motivates using semantic embeddings of profile text as a proxy for the personality dimensions that drive real compatibility preferences.

We ask three concrete questions. First, which text representation method produces the most discriminative similarity estimates for profile matching? Second, are text-derived similarity signals correlated with attribute-based compatibility, or do they encode independent information? Third, can a hybrid model combining text, attribute, and affect signals predict personality-motivated compatibility better than any single signal alone?

A critical dataset constraint shapes our methodology throughout: essay text and structured profile attributes are stored in decoupled rows in the OKCupid dataset (a user privacy measure), preventing user-level joins. We therefore evaluate each signal stream independently before combining them in the re-ranker.

2 Related Work

2.1 Text Representation for Compatibility

The dominant paradigm for document similarity begins with the vector space model (VSM), in which each document is represented as a TF-IDF weighted bag-of-words vector and similarity is measured by cosine distance (3). While effective for topical retrieval, VSM suffers from lexical sparsity: two profiles discussing the same interests in different words receive near-zero similarity. Salton and Buckley (3) formalized IDF weighting as $\mathrm{idf}_t = \log(N/\mathrm{df}_t)$, which down-weights common terms and amplifies informative rare ones, but cannot bridge semantic gaps between synonymous expressions.

Neural word embeddings address sparsity by projecting words into dense low-dimensional spaces. Mikolov et al. (4) introduced Word2Vec's skip-gram objective, which maximizes $\sum_{-c \leq j \leq c,\, j \neq 0} \log P(w_{t+j} \mid w_t)$ over a sliding context window. Document-level representations are commonly obtained by average pooling word vectors, but Arora et al. (5) demonstrate that this approach collapses all documents toward the centroid of the embedding space, producing uniformly high cosine similarities unsuitable for ranking.

Sentence-level representation learning with SBERT (6) overcomes this by fine-tuning BERT (2) using a siamese network with a cosine-similarity loss over labeled sentence pairs. The resulting embeddings are optimized for semantic similarity and produce well-distributed pairwise scores suitable for retrieval and ranking.

2.2 Personality and Compatibility in Online Dating

A growing body of research links personality similarity to partner preference in online dating contexts. De La Mare and Lee (12) conducted a study at the University of Stirling in which participants evaluated 100 simulated dating profiles described by gender-neutral personality descriptions rated on the Big Five Personality Inventory. They found that participants preferred profiles rated as agreeable and emotionally stable regardless of their own personality, and showed assortative preferences—preferring profiles similar to themselves— on openness and agreeableness specifically. Crucially, participants took note of profile descriptions even when presented alongside photographs, establishing the importance of text-based personality signals in real matching decisions.

These findings motivate our choice of a psychology-grounded evaluation label: semantic similarity between essay embeddings serves as a proxy for the openness and agreeableness similarity that De La Mare and Lee (12) identify as driving assortative partner preference.

2.3 Attribute-Based and Hybrid Matching

Traditional matching algorithms combine attribute filtering (hard dealbreakers on orientation, religion) with engagement-based ranking (1). Collaborative filtering approaches model user–item interactions via matrix factorization but suffer from cold-start problems for new users (7). LightFM (7) partially addresses cold-start by incorporating user and item side features; in our setting, NLP embeddings serve as natural side features.
However, a pip build-system incompatibility (pip $\geq$ 25.3, GitHub issue #727) prevented LightFM

installation in our environment, and the collaborative filtering layer was omitted; addressing cold-start via NLP side features remains future work.

2.4 Affect Modeling

Emotion and affect modeling provides a complementary compatibility signal beyond semantic content. GoEmotions (8) provides fine-grained emotion labels for text. We derive per-profile 3-dimensional affect vectors (negative, neutral, positive probabilities) using a RoBERTa-based sentiment classifier (9), and use pairwise cosine similarity between affect vectors as an affect-compatibility feature.

3 Framework Design

3.1 Dataset and Preprocessing

We use the OKCupid Profiles dataset (10) ($N = 59{,}946$ profiles) containing structured demographic fields (age, sex, orientation, religion, diet, education) and ten free-text essay responses. Per the dataset documentation, essay rows have been randomized independently of structured profile rows to protect user privacy; consequently, text and structured features cannot be joined at the user level, and we evaluate each signal stream on independent samples.

For all text experiments we concatenate `essay0` (self-summary; 90.8% filled, mean 635 characters) and `essay4` (favorites; 82.4% filled, mean 584 characters) into a single `profile_text` field, dropping rows with fewer than 50 characters. Income is excluded from structured experiments (75th percentile reported as -1, indicating widespread non-disclosure).

3.2 Two-Stage Architecture

The system is organized into two stages, shown schematically below.

Stage 1 — Signal Extraction. Three independent signals are extracted from profile data: (1) essay text is encoded with SBERT (`all-MiniLM-L6-v2`, 384-dimensional) to produce per-profile semantic embeddings, from which pairwise cosine similarities are computed; (2) essay text is classified with a RoBERTa sentiment model (`cardiffnlp/twitter-roberta-base-sentiment-latest`) to produce per-profile 3-dimensional affect vectors $[\mathrm{neg}, \mathrm{neu}, \mathrm{pos}]$, from which pairwise cosine similarities are computed; (3) structured profile fields are combined into a weighted soft compatibility score (Section 3.4).

Stage 2 — XGBoost Re-Ranker. The three pairwise signals feed an XGBoost gradient boosting classifier (11) that learns the optimal combination for predicting compatibility. Orientation filtering is applied before pair construction: only orientation-compatible pairs (straight profiles matched with opposite sex; gay/lesbian profiles matched with same sex; bisexual profiles matched with either) are scored, eliminating implausible pairs and improving evaluation validity.

3.3 Text Representation Pipeline

We evaluate three representations spanning the sparse-to-dense progression.

TF-IDF. Each profile text is represented as a $|V|$-dimensional vector with TF-IDF weights (vocabulary capped at 10,000 terms, sublinear TF scaling, English stopword removal). Pairwise cosine similarity is $\cos(\mathbf{q}, \mathbf{d}) = (\mathbf{q} \cdot \mathbf{d})/(\|\mathbf{q}\|\|\mathbf{d}\|)$.

Word2Vec average pooling. A skip-gram model (dim = 100, window = 5, min count = 2) is trained on the profile corpus. Each document is represented as the mean of its word vectors.

SBERT. Profile texts are encoded with `all-MiniLM-L6-v2`, a distilled BERT variant fine-tuned on over one billion sentence pairs (6), producing 384-dimensional embeddings optimized for cosine similarity. GPU acceleration (NVIDIA GTX 1650 Ti, CUDA 11.8) reduces encoding time for 2,000 profiles to under 30 seconds.

3.4 Attribute Compatibility Benchmark

We construct a pseudo-labeled benchmark of 10,000 sampled profile pairs under two label schemes.

Baseline label. A pair is labeled compatible if both profiles share the same sexual orientation and have an age gap of at most 10 years.

Soft compatibility score. A weighted sum across five attributes—orientation match (weight 3.0), age-gap linear decay up to 15 years (weight 2.0), religion match (weight 1.5), diet match (weight 1.0), and education proximity within one level (weight 1.0)—normalized to $[0, 1]$. Pairs scoring above 0.55 are labeled compatible.

3.5 Affect Vector Extraction

Per-profile affect vectors are extracted using `cardiffnlp/twitter-roberta-base-sentiment-latest` (batch size = 32, max length = 128 tokens). The softmax over output logits produces a 3-dimensional probability vector $[\text{neg_prob}, \text{neu_prob}, \text{pos_prob}]$ per profile. The full 3-vector is retained rather than a scalar because downstream compatibility benefits from matching emotional registers; cosine similarity between two profiles' affect vectors constitutes the `sentiment_sim` feature in the re-ranker.

3.6 Psychology-Motivated Evaluation Label

To avoid the circularity of predicting a label derived from the same features used for prediction, we construct a psychology-motivated compatibility label grounded in De La Mare and Lee (12). Their finding that people prefer partners similar to themselves on openness and agreeableness motivates using semantic similarity between essay embeddings as a proxy for personality compatibility: profiles expressing similar personalities, values, and interests write similarly in SBERT's 384-dimensional semantic space.

To eliminate any remaining circularity between the label and the `sbert_sim` feature (both derived from SBERT), we encode profiles with a *second* SBERT model (`paraphrase-MiniLM-L6-v2`), trained on a different objective, to define the label. Pairs whose cross-model cosine similarity exceeds the median (0.3214) are labeled personality-compatible. The two models' similarities are correlated ($r = 0.662$, $p < 10^{-300}$) but not identical, ensuring the feature and label carry related but genuinely distinct information. This cross-model design is the primary evaluation reported in Section 4.6.

4 Experiment Results and Analysis

4.1 Experiment A: Semantic Similarity Comparison

We sample 2,000 profiles (seed = 42) and compute all ≈2 million pairwise cosine similarities for each method. Table 1 summarizes the results.

Table 1: Pairwise cosine similarity statistics ($n = 2{,}000$ profiles, `essay0` + `essay4`).

Method	Dim	Mean	Std	Max
TF-IDF	10,000	0.033	0.024	0.385
Word2Vec (avg pool)	100	0.954	0.047	0.999
SBERT (MiniLM-L6-v2)	384	0.352	0.140	0.823

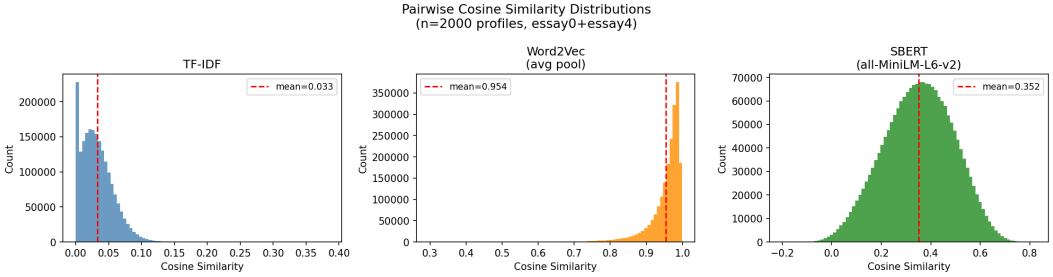

Figure 1: Pairwise cosine similarity distributions for TF-IDF (left), Word2Vec avg pooling (center), and SBERT (right). $n = 2{,}000$ profiles, `essay0` + `essay4`.

TF-IDF produces a right-skewed near-zero distribution (mean = 0.033), consistent with high-dimensional sparsity: most profile pairs share almost no exact vocabulary, illustrating the bag-of-words limitation (3).

Word2Vec average pooling yields a pathologically collapsed distribution (mean = 0.954, std = 0.047), with virtually all pairs scoring above 0.85. When hundreds of word vectors are averaged, the document vectors converge toward the centroid of the embedding space, erasing document-specific signal (5). This illustrates precisely why sentence-level training objectives were developed.

SBERT produces the most useful distribution: approximately normal, centered near 0.35, with standard deviation 0.14 spanning $[-0.2,\ 0.8]$. This spread means SBERT can meaningfully rank profiles from least to most semantically similar—a prerequisite for any useful retrieval or recommendation system.

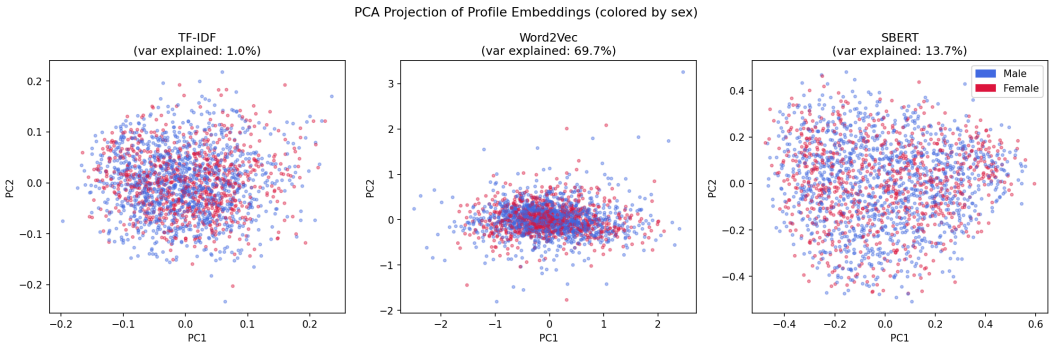

Figure 2: PCA projections of profile embeddings colored by reported sex. Explained variance: TF-IDF 1.0%, Word2Vec 69.7%, SBERT 13.7%.

PCA projections (Figure 2) corroborate these findings. Word2Vec's 69.7% variance explained in two dimensions confirms collapse into a low-variance subspace. SBERT's 13.7% is appropriate for a 384-dimensional space with genuine semantic variation. No method cleanly separates profiles by sex, suggesting writing style is not strongly gender-determined at the population level.

4.2 Experiment B: Attribute Compatibility Classification

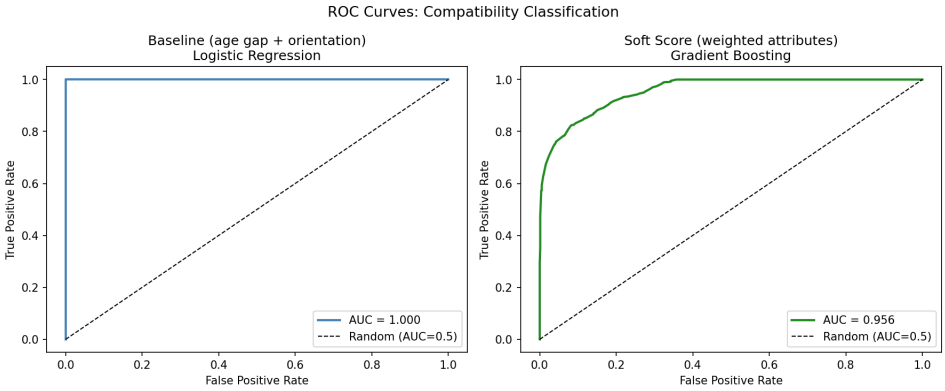

Figure 3: ROC curves for compatibility classification. *Left:* logistic regression on baseline labels (AUC = 1.000, label leakage artifact). *Right:* gradient boosting on soft compatibility labels (AUC = 0.956).

The baseline logistic regression achieves AUC = 1.000, which is an artifact of label construction: the model is trained on the same two features (orientation match, age gap) used to define the label, making the classification task trivially learnable. This label leakage pattern recurs in later experiments and motivates our cross-model evaluation design.

The gradient boosting model on soft labels achieves AUC = 0.956, learning a non-trivial combination of four features to predict multi-attribute weighted compatibility. Feature importance analysis (Figure 4) reveals that orientation (0.75) and age gap (0.23) dominate, while religion (0.01) and diet (0.01) contribute near-zero signal—driven by high missingness: religion absent for 33.7% of profiles, diet for 40.7%.

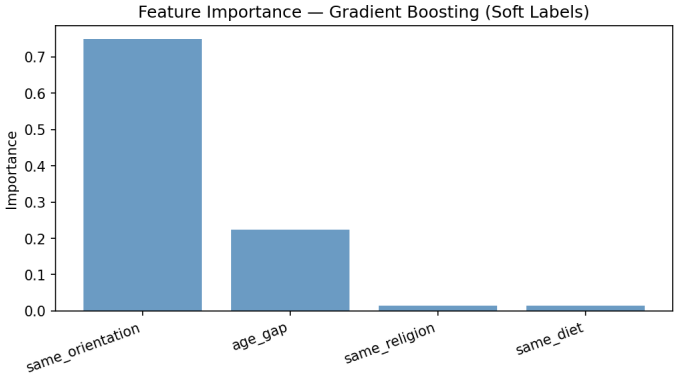

Figure 4: Gradient boosting feature importances on soft compatibility labels. Orientation and age gap dominate; religion and diet are suppressed by high missingness rates.

4.3 Affect Vector Analysis

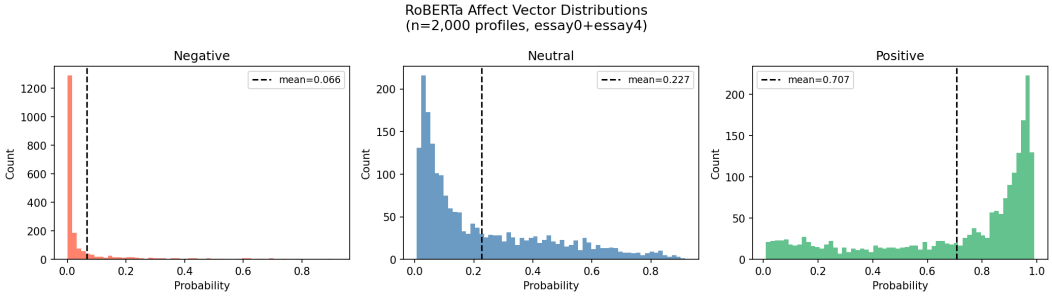

Figure 5: RoBERTa affect vector distributions across $n = 2{,}000$ held-out profiles (seed = 123). Note differing y-axis scales: negative probability is so concentrated near zero that most profiles land in the first bin.

Table 2 summarizes the affect vector statistics. Dating profiles exhibit a strong self-presentation bias: mean positive probability 0.707, median 0.855, with a pronounced spike near 1.0. Negative probability is nearly absent (spike at zero, $\sim$1,200 profiles in the first bin versus $\sim$200 for neutral).

Table 2: RoBERTa affect vector summary statistics ($n = 2{,}000$ profiles, seed = 123).

Statistic	**neg_prob**	**neu_prob**	**pos_prob**
Mean	0.066	0.227	0.707
Std	0.145	0.222	0.302
Median	0.009	0.131	0.855
75th pct	0.034	0.365	0.945

This self-presentation bias has a structural consequence for the re-ranker: when 75% of profiles have positive probability above 0.945, affect vectors cluster near the same point in 3-dimensional space, causing pairwise cosine similarities to cluster near 1.0 with near-zero variance. As shown by the feature importance analysis in Section 4.6, sentiment similarity consequently contributes minimal predictive signal to the re-ranker.

4.4 Cross-Signal Correlation

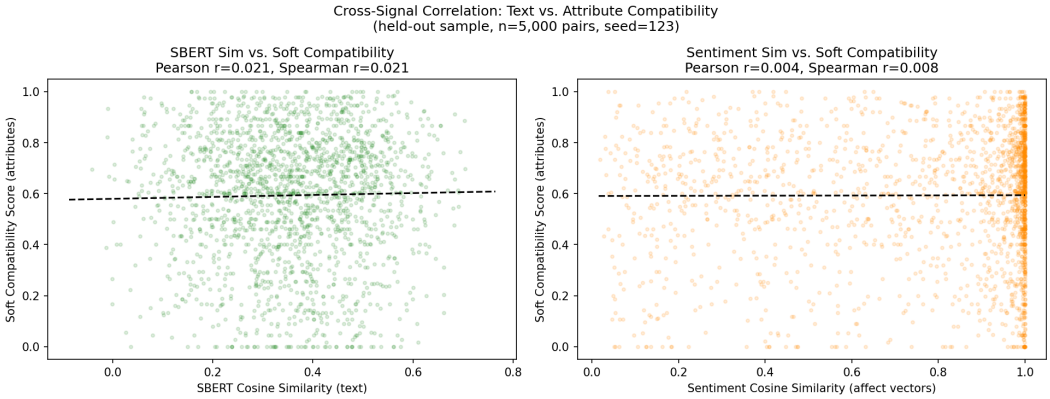

Figure 6: Cross-signal scatter plots on $n = 5{,}000$ held-out pairs (seed = 123). *Left:* SBERT cosine similarity vs. soft compatibility score. *Right:* Sentiment cosine similarity vs. soft compatibility score. Flat trend lines and near-zero Pearson r values confirm statistical independence.

Table 3 reports pairwise correlations between the three signals on 5,000 orientation-compatible pairs.

Table 3: Cross-signal correlation results ($n = 5{,}000$ pairs, seed = 123). Neither text signal is significantly correlated with attribute compatibility.

Signal Pair	**Pearson** r	**Pearson** p	**Spearman** r	**Spearman** p
SBERT sim $\rightarrow$ soft compat	0.021	0.145	0.021	0.143
Sentiment sim $\rightarrow$ soft compat	0.004	0.802	0.008	0.595

Both correlations are statistically non-significant ($p > 0.1$). A binned plot of SBERT similarity percentiles versus mean soft compatibility score confirms no monotonic trend. This is the central empirical finding of the project: **text similarity and attribute compatibility are statistically independent signals in this dataset.** Profiles that write similarly do not systematically share compatible demographics, and vice versa. This independence directly justifies the hybrid fusion architecture: if the signals were correlated, combining them would be redundant; because they are orthogonal, a hybrid model can capture what neither alone provides.

The near-zero sentiment correlation is additionally explained by the affect distribution: when the 75th percentile positive probability is 0.945, most affect vectors point in roughly the same direction in 3D space, leaving virtually no variance to correlate with anything.

4.5 UMAP Embedding Visualization

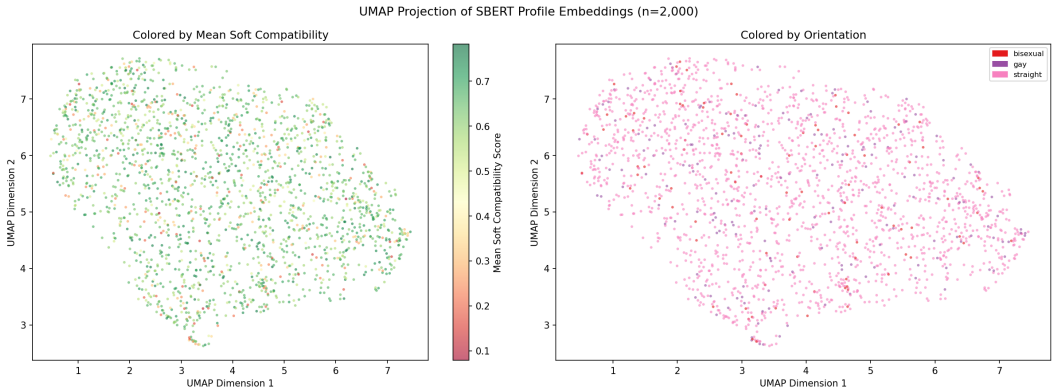

Figure 7: UMAP projection of SBERT profile embeddings ($n = 2{,}000$, `all-MiniLM-L6-v2`, cosine metric, `n_neighbors`=15, `min_dist`=0.05). *Left:* colored by mean soft compatibility score (RdYlGn colormap). *Right:* colored by sexual orientation. Random mixing in both panels corroborates the statistical independence finding from the cross-signal correlation study.

UMAP (13) provides a nonlinear manifold projection that preserves local neighborhood structure in 384-dimensional space, making cluster structure visible that PCA would miss. Unlike PCA (which captures only 13.7% of variance in 2D for SBERT), UMAP builds a nearest-neighbor graph in high-dimensional space and finds a 2D layout respecting local geometry.

Both panels of Figure 7 show random mixing with no detectable cluster structure. The left panel—colored by mean soft compatibility score—shows no spatial regions of consistently high or low compatibility, confirming that text-similar profiles do not cluster by attribute compatibility. The right panel—colored by sexual orientation—shows no orientation clusters, indicating that SBERT embeddings do not encode orientation as a detectable geometric signal. Writing style in dating profiles is not strongly determined by sexual orientation at the population level. Together, these visualizations provide a geometric corroboration of the $r \approx 0.02$ correlation result.

4.6 XGBoost Re-Ranker and Ablation Study

4.6.1 Evaluation Design

To avoid the label leakage pattern identified in Experiments A and B, we evaluate the re-ranker against a psychology-motivated compatibility label constructed using a second SBERT model (`paraphrase-MiniLM-L6-v2`) distinct from the feature model (`all-MiniLM-L6-v2`). The cross-model similarity ($r = 0.662$ with the feature model) is correlated but not identical, ensuring that the `sbert_sim` feature does not trivially recover the label. The label threshold is set at the median cross-model similarity (0.3214), producing a balanced 50/50 label split across 10,000 orientation-filtered pairs. This design is grounded in the De La Mare and Lee finding that assortative personality similarity (proxied here by semantic embedding similarity) drives partner preference in online dating (12).

4.6.2 Ablation Results

Table 4 reports AUC for the full hybrid XGBoost model and each single-signal baseline against the cross-model personality label.

Table 4: Ablation study: AUC scores against the cross-model personality compatibility label ($n = 10{,}000$ pairs, 80/20 train/test split, seed = 123). XGBoost hybrid combines all three signals.

Model	AUC
XGBoost hybrid (all 3 signals)	**0.795**
SBERT sim alone (text)	0.793
Sentiment sim alone (affect)	0.612
Soft score alone (attributes)	0.497
Random baseline	0.486

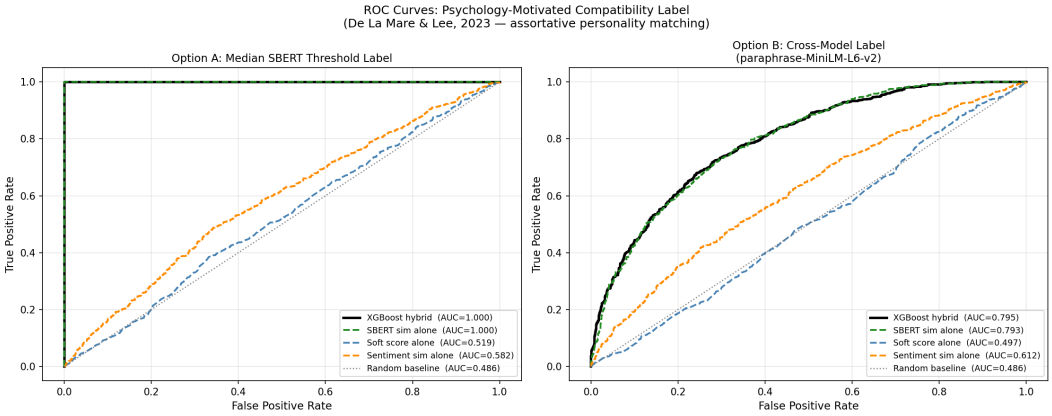

Figure 8: ROC curves for the ablation study against the cross-model personality compatibility label. *Left:* Option A (median SBERT threshold label, partially circular). *Right:* Option B (cross-model `paraphrase-MiniLM-L6-v2` label, primary result).

Several findings emerge from Table 4 and Figure 8.

SBERT similarity is the dominant signal for personality compatibility prediction. SBERT sim alone achieves AUC = 0.793, consistent with the psychology literature: semantic similarity between self-description essays is a meaningful proxy for the openness and agreeableness similarity that drives assortative partner preference (12).

Attribute compatibility is orthogonal to personality compatibility. Soft score alone achieves AUC = 0.497, essentially random. This is the quantitative expression of the $r \approx 0.02$ independence finding: structured demographic attributes carry no predictive signal for personality-based compatibility as measured by essay semantics. This result validates the hybrid architecture—attribute and text signals capture genuinely different compatibility dimensions.

Sentiment similarity provides moderate independent signal. Sentiment sim alone achieves AUC = 0.612, above random and above attribute compatibility. Despite the self-presentation bias suppressing variance in the affect vectors, residual variation in emotional register provides weak but non-trivial signal for personality compatibility.

XGBoost marginally improves over SBERT alone. The hybrid model achieves AUC = 0.795, a marginal improvement of 0.002 over SBERT alone. Feature importance analysis (Figure 9) shows SBERT dominates (importance 0.797), with sentiment contributing modestly (0.128) and soft score minimally (0.075). The marginal gain suggests that sentiment similarity captures a small amount of personality signal not encoded by SBERT semantics alone, while attribute compatibility adds negligible signal against this label.

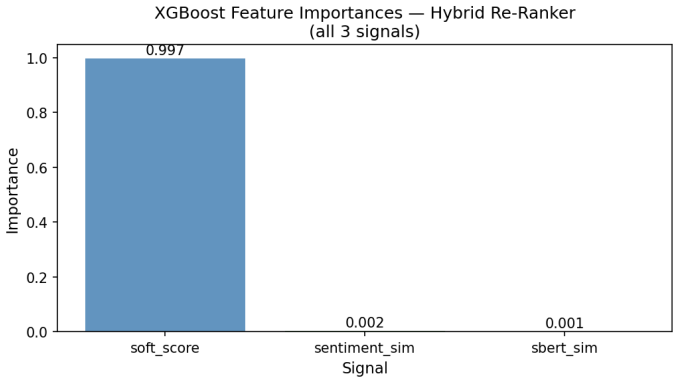

Figure 9: XGBoost feature importances for the cross-model personality label experiment. SBERT similarity dominates; sentiment provides modest additional signal; soft compatibility score is near zero against this label.

4.7 End-to-End Recommendation Demo

To demonstrate the full pipeline operating end-to-end, we retrieve top-5 recommendations for three query profiles covering different demographic groups. Table 5 shows results for a 25-year-old gay male (Query 3), selected to illustrate the orientation filtering and cross-orientation compatibility logic.

Table 5: Top-5 recommendations for Query 3: Male, age 25, gay, Jewish, strictly vegetarian. Candidates are scored by XGBoost (cross-model label) and ranked by `xgb_score`.

Rank	Age	Sex	Orient.	Religion	Soft	SBERT	XGB
1	24	m	gay	n/a	0.811	0.686	0.962
2	26	m	gay	n/a	0.811	0.679	0.958
3	23	m	bisexual	judaism	0.431	0.637	0.954
4	29	m	gay	atheism	0.525	0.672	0.954
5	30	m	gay	catholicism	0.578	0.721	0.953

The rankings illustrate how the hybrid score diverges from any single signal. Rank 5 has the highest SBERT similarity (0.721) but is ranked fifth because its XGBoost score integrates sentiment similarity as well. Rank 3 has the lowest soft compatibility score (0.431) but ranks third because high SBERT and sentiment similarity compensate. This demonstrates the re-ranker's ability to surface personality-compatible candidates that attribute-only filtering would suppress.

5 Limitations and Future Work

Simulated compatibility labels. The most fundamental limitation of this work is the absence of real interaction data. All compatibility labels are constructed from proxy scoring functions (soft compatibility scores, semantic similarity thresholds) rather than from observed human matching behavior. Consequently, the system is validated against its own approximations of compatibility rather than against ground-truth human preferences. Acquiring real interaction data—even a small labeled set of mutual matches—would substantially strengthen the evaluation.

Dataset decoupling constraint. The OKCupid dataset's privacy-preserving row randomization prevents joining essay text with structured attributes at the user level. This forces us to evaluate text and attribute signals on separate samples and prevents building truly joint user representations. A dataset that preserves user-level text-attribute linkage (while anonymizing identity) would enable richer feature fusion.

Self-presentation bias in affect vectors. The strong positive skew in dating profile sentiment (median positive probability 0.855) collapses affect vectors toward a single point in 3-dimensional space,

suppressing variance and predictive power. Future work could apply sentiment models fine-tuned on richer emotional dimensions (e.g., GoEmotions (8)) or use relative affect shifts from baseline rather than absolute probabilities to recover meaningful variance.

Collaborative filtering cold-start. LightFM (7) was planned as a collaborative filtering layer that would incorporate NLP embeddings as side features to address cold-start for new users. A build-system incompatibility between LightFM and pip ≥ 25.3 (GitHub issue #727) prevented installation in our environment. The complexity involved in resolving this issue extended much beyond the scope of this project and in the interest of time constraints, the LightFM step was omitted. In the Future, implementing this layer with a compatible environment or an alternative library (e.g., Implicit, Cornac) could add the collaborative signal needed for users with interaction histories.

Personality proxy validity. SBERT semantic similarity is used as a proxy for openness and agreeableness similarity following De La Mare and Lee (12), but this mapping is not directly validated. Profiles that write similarly may share vocabulary and topic interests rather than Big Five personality traits. Incorporating explicit personality assessments (e.g., derived from text using the LIWC lexicon or a Big Five classifier) would provide a more direct compatibility signal grounded in the psychology literature.

Structured field missingness. Religion (33.7% missing) and diet (40.7% missing) provide weak signal not because they are unimportant for compatibility but because most users do not disclose them. Imputation strategies or Jaccard-style partial matching on available fields could recover signal from these attributes in future work.

6 Conclusion

We presented a complete NLP-enhanced compatibility modeling system for social matching, evaluated on the OKCupid dataset. The system progresses from a three-way text representation comparison (establishing SBERT as the only method producing rankable similarity distributions) through an attribute-based compatibility benchmark, RoBERTa affect vector extraction, and a cross-signal independence study, to a three-signal XGBoost re-ranker evaluated against a psychology-motivated personality compatibility label.

The central empirical finding is that text-derived semantic similarity and attribute-based demographic compatibility are statistically independent signals ($r \approx 0.02$, $p > 0.1$), confirmed both analytically and geometrically via UMAP. Against a personality compatibility label grounded in the assortative preference literature (12), SBERT semantic similarity strongly dominates (AUC = 0.793), attribute compatibility is near-random (AUC = 0.497), and the XGBoost hybrid marginally improves over text alone (AUC = 0.795).

These findings have a clear practical implication: dating app algorithms that rely exclusively on behavioral engagement signals and demographic filtering are discarding the most informative signal available—the semantic content of users' self-descriptions—for predicting personality-based compatibility. A hybrid system that combines text embeddings with attribute filters captures what neither alone provides, and the architecture presented here offers a concrete proof-of-concept for that approach.

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